Can the existing Minas Gerais main grid deliver the 2035 and 2050 clean-energy portfolios?

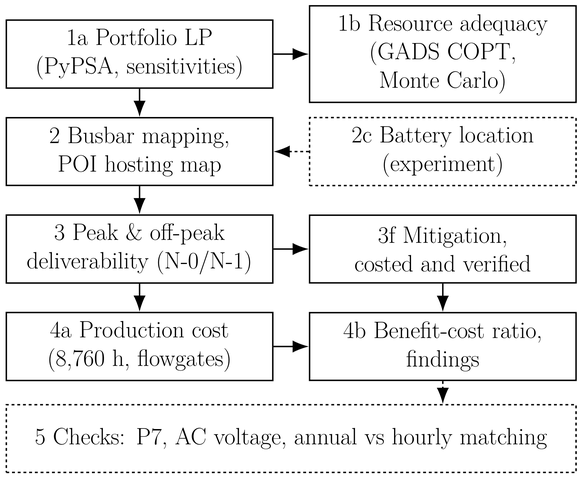
Xiao Huang Yunhan Huang Tianrun Yan Siyao Wu ENERGY 176/276 Electric System Planning, Stanford University

Question and scope

Following NREL's Oregon offshore-wind study: *how much new supply can the existing system host, and does the planned portfolio need major grid improvements?*

Minas Gerais (MG), Brazil, net zero by 2050 (PLAC-MG). Network: the MG main grid of the EPE 2035 planning case, 283 lines ≥230 kV and 272 transformer branches monitored, 437 N-1 contingencies. A facility is attributed to the portfolio inside the CAISO 5% circle (distribution factor ≥5% or flow impact ≥5% of its rating).

Approach (CAISO flow)



Solid: core assessment; dashed: experiment and checks. Forced outages: NERC GADS class averages. Costs: published EPE/ANEEL unit costs. Unserved energy: CCEE 2025 *custo do déficit*. Solvers: PyPSA with HiGHS.

Portfolios (new capacity, GW)

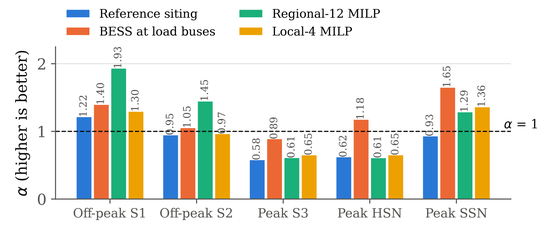
Year	PV	DG PV	Bio	SHP	Wind	Biogas	BESS	Total
2035	1.00	1.04	0.88	0.30	0.00	0.00	0.00	3.2
2050	12.92	7.94	0.88	0.67	2.00	0.40	5.29	30.1

PV central, DG distributed PV, Bio biomass, SHP small hydro, BESS 8-hour batteries. Hourly capacity-expansion LP: MG as one area, 8,760 hours, load 80.4 and 97.2 TWh. 2035 sits on the PLAC-MG policy floors; 2050 must be supplied locally without gas or imports. Our PyPSA rebuild reproduces both saved solutions.

Deliverability (N-0/N-1)

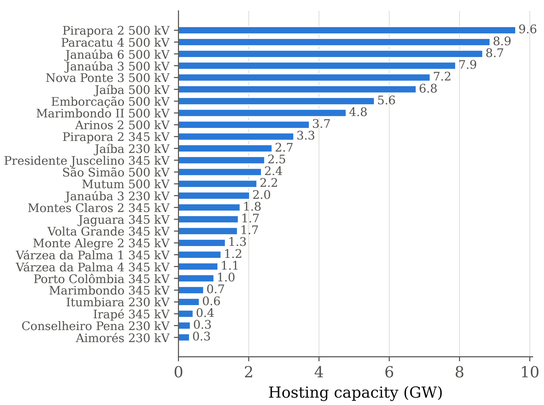
Study condition	2035 α ↑	2050 α ↑ Reference	2050 α ↑ BESS at load
S1 midday, after curtailment	7.48	1.22	1.40
S2 midday, before curtailment	7.34	0.95	1.05
S3 evening, batteries	5.58	0.58	0.89
HSN (CAISO)	5.89	0.62	1.18
SSN (CAISO)	7.01	0.93	1.65

↑ higher is better, ↓ lower is better; unmarked columns are sizes.



α is the largest scaling of the joint study output that meets every attributable limit; $\alpha < 1$ needs reinforcement. In 2050 the evening discharge of the 3.7 GW of northern batteries flows south through the Irapé, Montes Claros 2 and Janaúba 3 transformers: up to 9 upgrade groups.

Hosting map, 2050



Injection one candidate station can host alone before the first attributable violation (NREL Finding 1 style): 7 of 27 stations host at least 5 GW; the largest is Pirapora 2 (9,600 MW).

Resource adequacy

Year	Method, case	EUE (GWh) ↓	Firm gap (MW) ↓
2035	COPT (capacity only)	–	surplus 3,564
2035	Monte Carlo, normal	0.0	0
2035	Monte Carlo, dry	0.0	0
2050	COPT (capacity only)	–	surplus 1,283
2050	Monte Carlo, normal	44.0	≤94
2050	Monte Carlo, dry	2,859.3	≤1,594

↑ higher is better, ↓ lower is better; unmarked columns are sizes.

Method A: capacity outage probability table with NERC GADS EFORd by unit class. Method B: sequential Monte Carlo with two-state outage histories, each an 8,760-hour energy-limited PyPSA dispatch (monthly hydro budgets, storage); firm gaps by bisection (resolution 94 MW). **2050 is energy-limited, not capacity-limited:** the COPT finds a surplus, while the energy-limited dispatch falls short in every simulated year.

Mitigation and battery siting

2050 plan	Groups ↓	Capex (R$ M) ↓	BCR ↑
Reference siting, full plan	9	1,250	44.5
without small line needs	4	267	208.3
Batteries at load buses	1	47	55.0

↑ higher is better, ↓ lower is better; unmarked columns are sizes.

Alternatives per attributable group follow the PG&E cluster-study structure, costed from EPE/ANEEL unit costs, and the plan is verified on a rebuilt network that contains the new elements (smallest $\alpha = 1.000$). BCR = annual system-cost saving / annual revenue requirement; at VOLL −50% the full plan keeps BCR 22.5. Moving the batteries to load buses alone saves R$5,156 M/yr.

Checks

P7 (existing parallel circuits assumed to share towers): 2 more 230 kV circuits, R$65.1 M; full-plan BCR 42.3.

AC voltage, N-0: new out-of-limit buses: 2 high (2035), 1 low (2050 reference siting), up to 27 high with batteries at load (reactive switching not modelled).

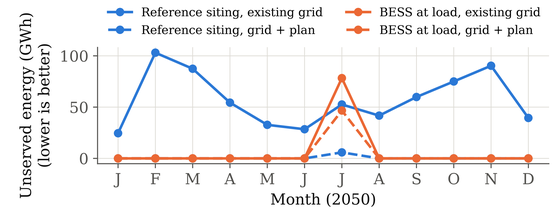
Annual vs hourly matching (2050): annual matching lowers cost by 48.7% if 16.6 TWh of exports can be delivered.

Limits: one evening network snapshot and one weather year; the 2050 network is the 2035 topology; DC flows.

Production cost, 2050

Network	Unserved (GWh) ↓	Congestion (R$ M/yr) ↓	Cong. curt. (%) ↓
Existing grid	689.5	5,824	0.33
Reference plan	5.9	54	0.01
BESS at load	78.3	667	0.06
BESS at load + plan	46.7	399	0.04

↑ higher is better, ↓ lower is better; unmarked columns are sizes.



8,760-hour PyPSA dispatch in monthly solves; flowgates are added lazily from all 242,708 N-0/N-1 rows until none is violated. With a plan, the remaining shortfalls fall in July, the month with the smallest hydro budget.

Findings

1. **2035 needs no major grid improvement:** α = 5.58–7.48, no congestion cost, adequate.
2. **2050 midday solar is deliverable** ($\alpha = 1.22$); **evening battery discharge is not** ($\alpha = 0.58$–0.62).
3. **Battery siting is the cheapest fix:** one bank (R$47 M) at load buses.
4. **With northern batteries the plan is justified:** 9 groups, R$267–1,250 M, BCR 44.5–208.3.
5. **2050 adequacy is limited by hydro energy, not capacity:** up to 94 MW (normal) and 1,594 MW (dry) of firm capacity.
6. **Curtailment comes from the hourly local-balance rule:** 15.01 TWh (13.4%) unused even without network limits; congestion adds 0.33%, below the 5% screen.
7. **The definition of fossil-free matters:** annual matching costs 48.7% less.

LaTeX TikZposter